QUESTION ANALYSIS

EEE459

Compound Semiconductor and Hetero-Junction Devices

Question Analysis

Sessions 2020–21 *1 papers, 22 questions, 48 formulas*

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1. At a glance:

This analysis covers **1 paper** (2020-21), with 22 questions and 150 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

With one paper, no trend can be read. Use this as a map of the topics.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Metal-semiconductor contacts	31	21 %	5	Barrier lowering (14); Schottky junction (13); Ohmic contacts (4)
JFETs	29	19 %	4	Pinch-off (15); Transconductance (8); Channel profiles (6)
Alloys and strain	20	13 %	3	Lattice matching (9); Quaternary alloys (7); Isoelectronic traps (4)
Heterojunctions	20	13 %	2	Fields and potentials (12); Band diagrams (8)
Band structure and materials	17	11 %	3	Hole bands (6); Density of states (6); Materials (5)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2020-21	All
Band structure and materials	**17**	**17**
Hole bands	**6**	6
Density of states	**6**	6
Materials	**5**	5
High-field transport	**11**	**11**
Mobility	**6**	6
Gunn effect	**5**	5
Alloys and strain	**20**	**20**
Lattice matching	**9**	9
Quaternary alloys	**7**	7
Isoelectronic traps	**4**	4
PN junction	**5**	**5**
Graded junction	**5**	5
Metal-semiconductor contacts	**31**	**31**
Barrier lowering	**14**	14
Schottky junction	**13**	13
Ohmic contacts	**4**	4
Heterojunctions	**20**	**20**
Fields and potentials	**12**	12
Band diagrams	**8**	8
Bipolar transistors	**17**	**17**
Ebers-Moll	**9**	9
High-level injection	**8**	8
JFETs	**29**	**29**

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Topic and subtopic	2020-21	All
Pinch-off	**15**	15
Transconductance	**8**	8
Channel profiles	**6**	6

3. Priority list:

Must know: Metal-semiconductor contacts (31), JFETs (29), Alloys and strain (20), Heterojunctions (20), Band structure and materials (17), Bipolar transistors (17), High-field transport (11), PN junction (5)

Likely: none

Less often: none

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. Band structure and materials:

17 marks, 3 questions, in 1 of 1 papers. Book: Neamen Ch. 3; Sze Ch. 1.

Most often theory (2 of 3). Subtopics: Hole bands; Materials; Density of states.

Formulas:

Subtopic	Formula	Equation	Used in
Hole bands (6)	**Parabolic band**	$E = \frac{\hbar^2 k^2}{2m^*}$	20-21 1(c)
	Slope and curvature	$\frac{dE}{dk} = \frac{\hbar^2 k}{m^*}, \quad \frac{d^2E}{dk^2} = \frac{\hbar^2}{m^*}$	20-21 1(c)
	Si hole masses	$m_{hh,lh}[100] = \frac{m_0}{\gamma_1 \mp 2\gamma_2}, \quad m_{hh,lh}[111] = \frac{m_0}{\gamma_1 \mp 2\gamma_3}$	20-21 1(c)
Materials (5)	**Emission wavelength**	$\lambda\ (\mu\text{m}) = \frac{1.24}{E_g\ (\text{eV})}$	20-21 2(a)
Density of states (6)	**Density of states, 3D**	$g_{3D}(E) = \frac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E - E_c}$	20-21 3(b)
	Density of states, 2D	$g_{2D}(E) = \frac{m^*}{\pi\hbar^2}$ per subband	20-21 3(b)
	Density of states, 1D	$g_{1D}(E) \propto \frac{1}{\sqrt{E - E_n}}$	20-21 3(b)

Symbols:

E = energy from the band edge (J) ↩

$\hbar$ = reduced Planck constant (J · s) ↩

g = states per unit energy per unit volume

k = wave number (m^{-1}) ↩

λ = wavelength of a photon of energy E_g ↩

E_c = conduction-band edge ↩

Questions:

No.	Paper	Marks	Type	Question
Hole bands				
1	20-21 1(c)	6	derivation	Prove that along the [100] and [111] direction in k-space, for Si, the variation of electron energy with respect to wave number is higher for light holes than for heavy holes.
Materials				
2	20-21 2(a)	5	theory	Which semiconductors are most extensively used in near infra-red region? Explain the reasons behind it.
Density of states				
3	20-21 3(b)	6	theory	Explain how the density of states changes qualitatively as the dimensionality reduces from $3D \to 2D \to 1D$. Sketch $D(E)$ vs E for each case.

4.2. High-field transport:

11 marks, 2 questions, in 1 of 1 papers. Book: Sze Ch. 1, 10.

Most often theory (2 of 2). Subtopics: Mobility; Gunn effect.

Formulas:

Subtopic	Formula	Equation	Used in
Mobility (6)	**Mobility**	$\mu = \dfrac{e\,\tau}{m^*}$	20-21 1(a)
Gunn effect (5)	Drift velocity, low field	$v_d = \mu E$	20-21 1(b)
	Saturation velocity	$v_d \to v_{sat} \approx 10^7$ cm/s	20-21 1(b)
	Gunn oscillation	$f = \dfrac{v_d}{L}$	20-21 1(b)

Symbols:

μ = mobility ($\mathrm{cm}^2/\mathrm{V}\cdot\mathrm{s}$) ↩

τ = mean time between collisions (s) ↩

m^* = conductivity effective mass (kg) ↩

v_d = drift velocity (cm/s) ↩

E = electric field (V/cm) ↩

f = transit-time frequency (Hz) ↩

L = length of the GaAs sample (cm) ↩

Questions:

No.	Paper	Marks	Type	Question
Mobility				
1	20-21 1(a)	6	theory	Between Si and GaAs, which has higher electron mobility and why?
Gunn effect				
2	20-21 1(b)	5	theory	Discuss Velocity saturation and Gunn effect .

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4.3. Alloys and strain:

20 marks, 3 questions, in 1 of 1 papers. Book: Neamen Ch. 14.

Most often theory (2 of 3). Subtopics: Lattice matching; Isoelectronic traps; Quaternary alloys.

Formulas:

Subtopic	Formula	Equation	Used in
Lattice matching (9)	**Lattice misfit**	$f = \dfrac{a_{layer} - a_{sub}}{a_{sub}}$	20-21 2(b)
	Critical thickness, estimate	$h_c \sim \dfrac{a_{sub}}{2\lvert f\rvert}$	20-21 2(b)
Quaternary alloys (7)	**Vegard's law, ternary**	$a(A_xB_{1-x}C) = x\,a_{AC} + (1-x)\,a_{BC}$	20-21 2(d)
	Quaternary $In_xGa_{1-x}As_ySb_{1-y}$	$P = xy\,P_{InAs} + x(1-y)\,P_{InSb} + (1-x)y\,P_{GaAs} + (1-x)(1-y)\,P_{GaSb}$	20-21 2(d)

Symbols:

a_{layer} = layer lattice constant ↩

a_{sub} = substrate lattice constant ↩

a = lattice constant (Å) ↩

x = mole fraction ↩

P = lattice constant or band gap of the alloy, from the four binaries ↩

Questions:

No.	Paper	Marks	Type	Question
Lattice matching				
1	20-21 2(b)	9	theory	Why is lattice matching of semiconductor important for hetero-structured devices? Discuss the effect of strain and critical thickness in epitaxial growth.
Isoelectronic traps				
2	20-21 2(c)	4	theory	Discuss the effect of isoelectric trap in the ternary compound semiconductor $GaAs_{1-x}P_x$.
Quaternary alloys				
3	20-21 2(d)	7	numerical	Fig. 1 shows the Bandgap Vs. lattice constant for different binary compounds. Assume that a quaternary compound $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. Find the composition if the bandgap is 1 eV and lattice parameter is 5.93 Å.

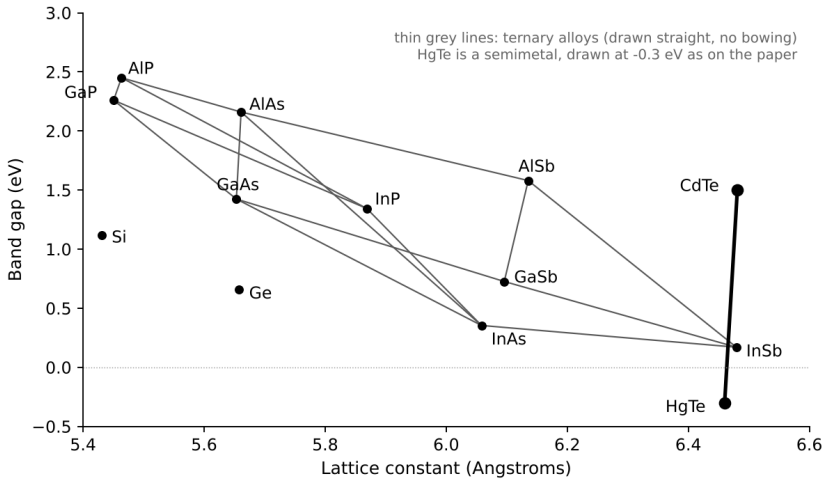


Figure 1: 20-21 2(d): Fig. 1: Band gap vs. lattice constant of binary compounds (redrawn from the paper)

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4.4. PN junction:

5 marks, 1 question, in 1 of 1 papers. Book: Neamen Ch. 7.

Most often theory (1 of 1). Subtopics: Graded junction.

Formulas:

Subtopic	Formula	Equation	Used in
Graded junction (5)	Net doping	$N_d - N_a = a\,x$	20-21 4(c)
	Depletion width	$W = \left[\frac{12\epsilon_s(V_{bi}+V_R)}{ea}\right]^{1/3}$	20-21 4(c)
	Junction capacitance	$C' = \left[\frac{ea\epsilon_s^2}{12(V_{bi}+V_R)}\right]^{1/3}$	20-21 4(c)

Symbols:

a = grading constant (cm^{-4}) ↩ W = total depletion width (cm) ↩

Questions:

No.	Paper	Marks	Type	Question
Graded junction				
1	20-21 4(c)	5	theory	Describe how a linearly graded junction is formed.

4.5. Metal-semiconductor contacts:

31 marks, 5 questions, in 1 of 1 papers. Book: Neamen Ch. 9; Sze Ch. 3.

Most often theory (3 of 5). Subtopics: Schottky junction; Barrier lowering; Ohmic contacts.

Formulas:

Subtopic	Formula	Equation	Used in
Schottky junction (13)	**Barrier height (ideal)**	$\phi_{B0} = \phi_m - \chi$	20-21 3(c)
	Built-in potential	$V_{bi} = \phi_{B0} - \phi_n,\ \phi_n = \frac{kT}{e}\ln\frac{N_c}{N_d}$	20-21 3(c)
	Depletion width	$x_n = \left[\frac{2\epsilon_s(V_{bi}+V_R)}{eN_d}\right]^{1/2}$	20-21 3(c)
	Maximum field	$E_{max} = \frac{eN_dx_n}{\epsilon_s} = \frac{2(V_{bi}+V_R)}{x_n}$	20-21 3(c)
	Junction capacitance	$C' = \frac{\epsilon_s}{x_n} = \left[\frac{e\epsilon_sN_d}{2(V_{bi}+V_R)}\right]^{1/2}$	20-21 3(c)
Barrier lowering (14)	**Image-force lowering**	$\Delta\phi = \sqrt{\frac{eE}{4\pi\epsilon_s}}$	20-21 3(d)
	Position of the peak	$x_m = \sqrt{\frac{e}{16\pi\epsilon_sE}}$	20-21 3(d)
	Barrier with interface states	$\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)\left(\frac{E_g}{e} - \phi_0\right)$	20-21 4(b)
Ohmic contacts (4)	Ohmic contact (ideal)	$\phi_m < \phi_s$ (n-type), $\phi_m > \phi_s$ (p-type)	20-21 5(d)

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Symbols:

ϕ_m = metal work function (V) ↩

χ = electron affinity (V) ↩

ϕ_n = depth of E_F below E_c (V) ↩

N_c = conduction-band density of states (cm^{-3}) ↩

V_R = reverse bias (V) ↩

C' = capacitance per unit area (F/cm^2) ↩

$\Delta\phi$ = barrier lowering (V) ↩

E = field at the interface (V/cm)

x_m = distance of the barrier maximum from the metal (cm) ↩

$\gamma = 1$ without interface states, 0 when the Fermi level is pinned ↩

ϕ_0 = neutral level of the interface states above E_v (V) ↩

ϕ_s = semiconductor work function (V) ↩

Questions:

No.	Paper	Marks	Type	Question
Schottky junction				
1	20-21 3(a)	5	theory	Draw the ideal energy band diagram of a metal semiconductor junction under: (i) Reverse bias condition (ii) Forward bias condition
2	20-21 3(c)	8	numerical	Consider a contact between tungsten and n-type silicon doped to $N_d = 1 \times 10^{16}$ cm^{-3} at $T = 300$ K. Determine the (i) theoretical barrier height (ii) built-in potential barrier (iii) maximum electric field in the metal-semiconductor diode for zero applied bias, and (iv) the junction capacitance. Given that, the metal work function for tungsten (W) is 4.55 V and the electron affinity for silicon is 4.01 V. Assume $N_c = 2.8 \times 10^{19}$ cm^{-3}.
Barrier lowering				
3	20-21 3(d)	6	numerical	Consider a gallium arsenide metal-semiconductor contact in which the electric field in the semiconductor is assumed to be $E = 6.8 \times 10^4$ V cm^{-1}. Calculate the Schottky barrier lowering (due to image force induced lowering) and the position of the maximum barrier height.
4	20-21 4(b)	8	theory	Discuss the following non-ideal effects on barrier height of a metal-semiconductor heterojunction: (i) Image-force-induced lowering (ii) Interface states
Ohmic contacts				
5	20-21 5(d)	4	theory	Sketch the ideal energy band diagram of a metal semiconductor junction in which $\Phi_m < \Phi_s$. Explain why this is an ohmic contact.

4.6. Heterojunctions:

20 marks, 2 questions, in 1 of 1 papers. Book: Neamen Ch. 9.

Most often derivation (2 of 2). Subtopics: Band diagrams; Fields and potentials.

Formulas:

Subtopic	Formula	Equation	Used in
Band diagrams (8)	AlGaAs band gap	$E_g(Al_xGa_{1-x}As) = 1.424 + 1.247x$ eV	20-21 1(d)
	Band offsets (Anderson)	$\Delta E_c = e(\chi_n - \chi_w),\ \Delta E_g = \Delta E_c + \Delta E_v$	20-21 1(d)
	Hole concentration	$p = N_v \exp\left[-\frac{E_F - E_v}{kT}\right]$	20-21 1(d)
	Built-in potential, pN	$eV_{bi} = -\Delta E_v + kT \ln\left(\frac{p_P\, N_{vN}}{p_N\, N_{vP}}\right)$	20-21 1(d)
Fields and potentials (12)	Field, N side	$E_1(x) = \frac{eN_{d1}}{\epsilon_1}(x + x_n)$	20-21 4(a)
	Field, P side	$E_2(x) = \frac{eN_{a2}}{\epsilon_2}(x_p - x)$	20-21 4(a)

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Subtopic	Formula	Equation	Used in
	Interface condition	$\epsilon_1 E_1(0) = \epsilon_2 E_2(0) \Rightarrow N_{d1} x_n = N_{a2} x_p$	20-21 4(a)
	Potential split	$V_{bi1} = \frac{eN_{d1}x_n^2}{2\epsilon_1},\ V_{bi2} = \frac{eN_{a2}x_p^2}{2\epsilon_2},\ V_{bi} = V_{bi1} + V_{bi2}$	20-21 4(a)
	Depletion width, N side	$x_n = \left[\frac{2\epsilon_1\epsilon_2 N_{a2} V_{bi}}{eN_{d1}(\epsilon_1 N_{d1} + \epsilon_2 N_{a2})}\right]^{1/2}$	20-21 4(a)

Symbols:

χ_n, χ_w = electron affinities of the narrow- and wide-gap materials (V) ↩

ΔE_c, ΔE_v = conduction- and valence-band offsets (eV) ↩

N_v = valence-band effective density of states (cm^{-3}) ↩

p_P, p_N = hole concentrations in the P and N regions ↩

N_{vP}, N_{vN} = their valence-band densities of states ↩

N_{d1} = donors ↩

ϵ_1 = permittivity of side 1 ↩

N_{a2} = acceptors ↩

ϵ_2 = permittivity of side 2 ↩

V_{bi1}, V_{bi2} = parts of V_{bi} dropped on each side ↩

Questions:

No.	Paper	Marks	Type	Question
Band diagrams				
1	20-21 1(d)	8	derivation	For a $GaAs - Al_xGa_{1-x}As$ heterojunction, determine the following: (i) Energy-Band diagram: Case-1: Before contact, Case-2: After contact (ii) Show that the built-in potential (using hole concentration) $eV_{bi} = -\Delta E_v + kT \ln\left(\frac{P_P N_{VN}}{P_N N_{VP}}\right)$. Given, GaAs is doped P type, $Al_xGa_{1-x}As$ is doped N^+ type. Assume: $x = 0.25$ & $\Delta E_c = \frac{1}{2}\Delta E_g$ P_p = hole concentration in P type region, P_N = hole concentration in N type region, N_{VN} and N_{VP} are density of states of electron and hole respectively.
Fields and potentials				
2	20-21 4(a)	12	derivation	A narrow bandgap N-type and a wide bandgap P-type material are used to form a heterojunction. If they are straddling type, then draw the energy band diagram of the heterojunction before contact and after contact at thermal equilibrium. With necessary assumption, derive expression for electric field and electrostatic potential in both regions.

4.7. Bipolar transistors:

17 marks, 2 questions, in 1 of 1 papers. Book: Neamen Ch. 12; Sze Ch. 5.

Most often derivation (1 of 2). Subtopics: Ebers-Moll; High-level injection.

Formulas:

Subtopic	Formula	Equation	Used in
Ebers-Moll (9)	**Ebers–Moll currents**	$I_C = \alpha_F I_F - I_R,\ I_E = I_F - \alpha_R I_R,\ I_B = I_E - I_C$	20-21 6(a)
	Diode currents	$I_F = I_{ES}\left[e^{V_{BE}/V_t} - 1\right],\ I_R = I_{CS}\left[e^{V_{BC}/V_t} - 1\right]$	20-21 6(a)
	Reciprocity	$\alpha_F I_{ES} = \alpha_R I_{CS}$	20-21 6(a)
	Saturation voltage	$V_{CE}(sat) = V_t \ln\left[\frac{I_C(1-\alpha_R) + I_B}{\alpha_F I_B - (1-\alpha_F) I_C} \cdot \frac{\alpha_F}{\alpha_R}\right]$	20-21 6(a)

Symbols:

α_F, α_R = forward and reverse common-base gains ↩

I_{ES}, I_{CS} = reverse saturation currents of the B–E and B–C junctions (A) ↩

Questions:

No.	Paper	Marks	Type	Question
Ebers-Moll				
1	20-21 6(a)	9	derivation	Draw the equivalent circuit for basic Ebers-Moll model as used in BJT. Also derive the expression of $V_{CE}(sat)$ from this model.
High-level injection				
2	20-21 6(b)	8	theory	Explain high level injection and current crowding effects in a BJT with necessary figures. How can the problem of current crowding be solved?

4.8. JFETs:

29 marks, 4 questions, in 1 of 1 papers. Book: Neamen Ch. 13; Sze Ch. 7.

Most often numerical (2 of 4). Subtopics: Channel profiles; Pinch-off; Transconductance.

Formulas:

Subtopic	Formula	Equation	Used in
Channel profiles (6)	Saturation drain voltage	$V_{DS}(sat) = V_{p0} - (V_{bi} - V_{GS})$	20-21 5(a)
Pinch-off (15)	pn built-in potential	$V_{bi} = V_t \ln \dfrac{N_a N_d}{n_i^2}$	20-21 5(c)
	Space-charge width	$h = \left[\dfrac{2\epsilon_s(V_{bi} - V_{GS})}{eN_d}\right]^{1/2}$	20-21 5(b)
	Internal pinch-off voltage	$V_{p0} = \dfrac{ea^2 N_d}{2\epsilon_s}$	20-21 5(b)
	Pinch-off voltage	$V_p = V_{bi} - V_{p0}$	20-21 5(b)
Transconductance (8)	**Pinch-off current**	$I_{P1} = \dfrac{\mu_n (eN_d)^2 W a^3}{6\epsilon_s L}$	20-21 6(c)
	Saturation current	$I_{D1}(sat) = I_{P1}\left[1 - 3\left(\dfrac{V_{bi} - V_{GS}}{V_{p0}}\right)\left(1 - \dfrac{2}{3}\sqrt{\dfrac{V_{bi} - V_{GS}}{V_{p0}}}\right)\right]$	20-21 6(c)
	Transconductance	$g_{ms} = \dfrac{3I_{P1}}{V_{p0}}\left(1 - \sqrt{\dfrac{V_{bi} - V_{GS}}{V_{p0}}}\right)$	20-21 6(c)

Symbols:

N_a = gate acceptors ↩

N_d = channel donors (cm^{-3}) ↩

h = depletion width into the channel (cm) ↩

V_{GS} = gate–source voltage (V) ↩

a = metallurgical channel thickness (cm)

W = channel width

L = channel length (cm)

μ_n = electron mobility ($\text{cm}^2/\text{V}\cdot\text{s}$) ↩

g_{ms} = saturation transconductance (S) ↩

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Questions:

No.	Paper	Marks	Type	Question
Channel profiles				
1	20-21 5(a)	6	theory	Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where V_p, V_g, and V_d are pinch-off, gate, and drain voltage respectively) (i) $V_g > V_p$ and $V_d < V_{d,sat}$ (ii) $V_g > V_p$ and $V_d = V_{d,sat}$ (iii) $V_g > V_p$ and $V_d > V_{d,sat}$
Pinch-off				
2	20-21 5(b)	8	derivation	Derive the formula of internal pinch-off voltage in an n-channel JFET.
3	20-21 5(c)	7	numerical	Consider a GaAs n-channel pn JFET at $T = 300$ K with $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 3 \times 10^{15}$ cm^{-3}, $\epsilon_s = 13.1$, and $a = 0.70$ μm. Determine the forward-bias gate voltage required to open a channel region that is 0.10 μm thick with zero drain voltage.
Transconductance				
4	20-21 6(c)	8	numerical	Consider a silicon n-channel JFET at $T = 300$ K with the following parameters: $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 1 \times 10^{16}$ cm^{-3}, $a = 0.75$ μm, $L = 10$ μm, $W = 30$ μm, and $\mu_n = 1000$ cm^2/(V s) (i) Calculate the maximum current in an n-channel JFET. (ii) Determine the maximum transconductance of an n-channel depletion mode JFET biased in the saturation region. $I_{P1} = 0.522$ mA, $V_{bi} = 0.814$ V, and $V_{po} = 4.35$ V.

5. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Band structure and materials				
1	20-21 1(c)	6	Hole bands	Differentiate the parabolic band; compare slopes; put in the effective masses.
Heterojunctions				
2	20-21 1(d)	8	Band diagrams	Write the hole concentrations; solve for the Fermi levels; subtract; use the valence-band offset.
3	20-21 4(a)	12	Fields and potentials	Integrate Poisson's equation on each side; apply D continuity; integrate again for the potential.
Bipolar transistors				
4	20-21 6(a)	9	Ebers-Moll	Write the terminal currents; solve for the diode currents; subtract the junction voltages.
JFETs				
5	20-21 5(b)	8	Pinch-off	Integrate Poisson's equation in the depletion layer; set the depletion width equal to the channel thickness.

6. Repeat tracker:

No repeats among these papers.

7. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical
Band structure and materials	2	1	·
High-field transport	2	·	·
Alloys and strain	2	·	1
PN junction	1	·	·
Metal-semiconductor contacts	3	·	2
Heterojunctions	·	2	·
Bipolar transistors	1	1	·
JFETs	1	1	2

8. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Metal-semiconductor contacts	31	21 %
2	JFETs	29	40 %
3	Alloys and strain	20	53 %
4	Heterojunctions	20	67 %
5	Band structure and materials	17	78 %
6	Bipolar transistors	17	89 %
7	High-field transport	11	97 %
8	PN junction	5	100 %

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